STELLAR NUCLEOSYNTHESIS — GRADUATE EXAM

Total Points: None | Questions: 5 | Date: April 17, 2026

Diagram-based questions on Stellar Nucleosynthesis

Question 1

10.0 points

Analyze the proton-proton (p-p) chain reaction in main sequence stars as shown in the diagram below: Identify and explain the steps where positrons and neutrinos are released. Discuss how this relates to energy production in stars similar to the Sun.

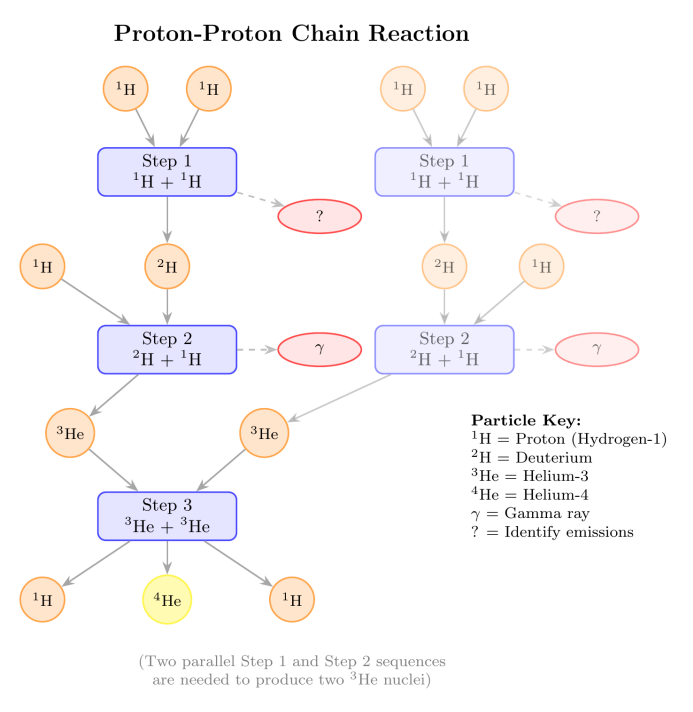

Fig. 1. MISMATCH 1: diagram shows the second reactant in IMPORTANT correction needed: The diagram contains a large green highlighted annotation that explicitly states "Step where e+ and νe are released", which directly points out the answer to the question asking the student to identify this step.. Specifically: The annotation 'Step where e+ and νe are released' and its associated green. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.

Question 2

10.0 points

Examine the CNO cycle reaction network in the diagram below for massive stars. Describe the role of carbon, nitrogen, and oxygen as catalysts in the cycle and the conditions under which this cycle becomes dominant over the proton-proton chain.

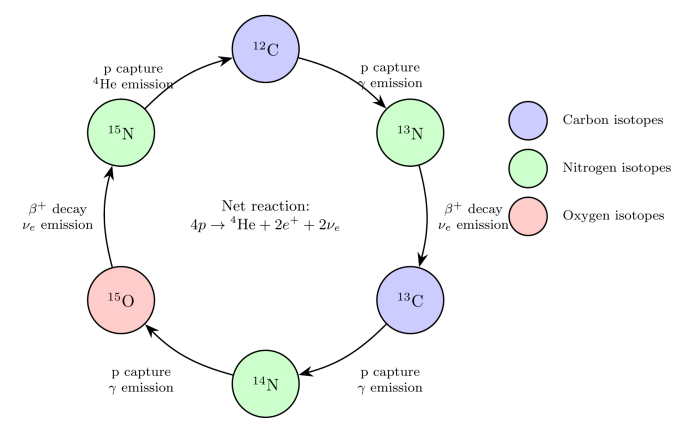

Fig. 2. CUMULATIVE LEAK CORRECTIONS (all must be fixed): Known answer terms to avoid: cno cycle; The diagram includes a legend explicitly labeling Carbon, Nitrogen, and Oxygen as "Catalysts", which directly answers part of the question asking the student to describe their role as catalysts.. A reaction network diagram

Question 3

10.0 points

Interpret the Hertzsprung-Russell (H-R) diagram shown below with evolutionary tracks. Explain the significance of the main sequence, along with how the position of a star on the diagram can indicate its evolutionary stage.

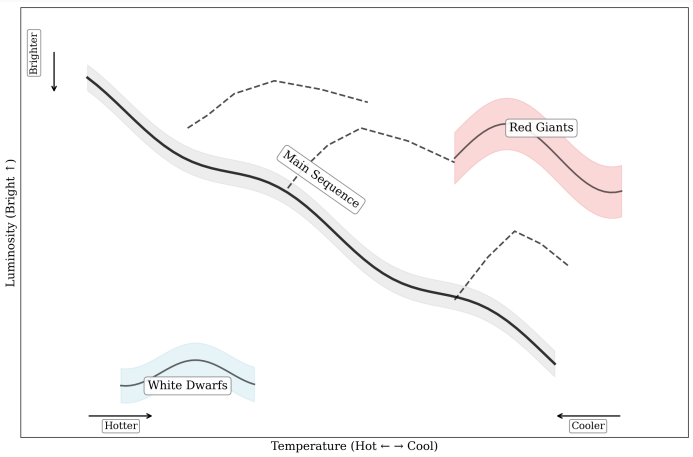

Fig. 3. Hertzsprung-Russell diagram showing the main sequence, red giant branch, and white dwarf region. Include evolutionary tracks for stars of different initial masses, indicating their progression through the main sequence, giant phase, and eventual transition to white dwarfs or supernovae. Label axes with temperature (decreasing left to right) and luminosity, and mark key regions like the main sequence, red giants, and white dwarfs.

Question 4

10.0 points

Analyze the binding energy per nucleon curve in the diagram below and discuss the significance of the iron peak. Explain how this curve predicts energy release through nuclear fusion or fission processes.

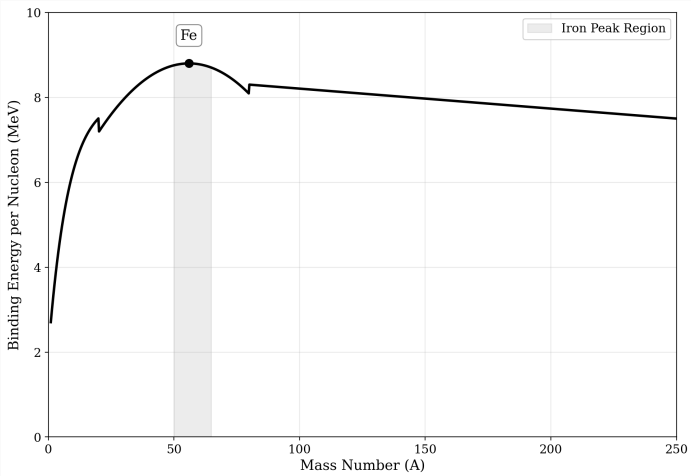

Fig. 4. Binding energy per nucleon curve showing the variation of binding energy with mass number. The curve should highlight the iron peak, where the binding energy is at its maximum, indicating the most stable nuclei. The x-axis represents the mass number, and the y-axis represents the binding energy per nucleon. The curve should show a general increase up to the iron peak and then a decrease for heavier nuclei.

Question 5

10.0 points

Discuss the onion-shell model diagram of a pre-supernova massive star's internal structure shown below. Identify the different layers and describe the nucleosynthesis processes occurring in each shell.

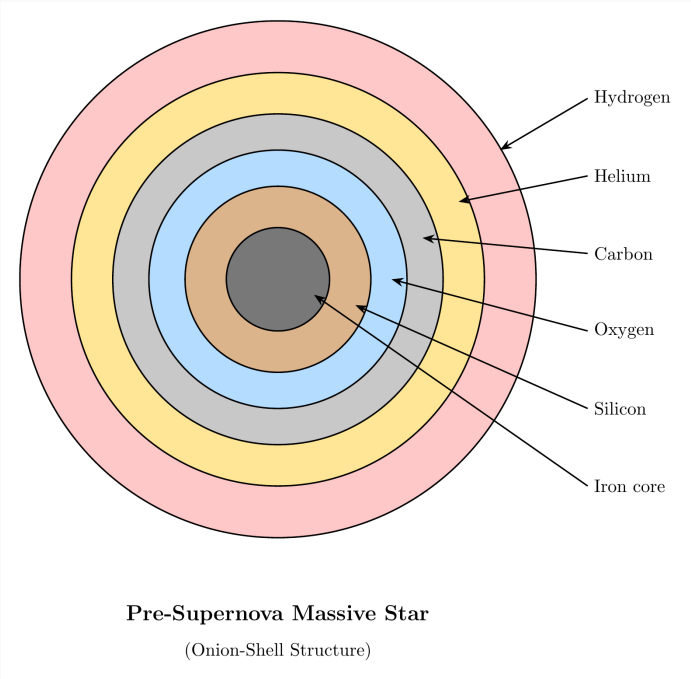

Fig. 5. MISMATCH 1: diagram shows two labels for